Lara María Vázquez González

Bioinformatics · NGS Data Analysis · Machine Learning

CONTACT

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Santiago de Compostela

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LANGUAGES

English	C2 (Cambridge)
Spanish	Native
Galician	Native

EDUCATION

PhD in Information Technology Research
Universidade de Santiago de Compostela
2020 – 2025

MSc in Bioinformatics and Biostatistics
Universitat Oberta de Catalunya
2019 – 2020

BSc in Computer Engineering
Universidade de Santiago de Compostela
2015 – 2019

PROGRAMMING

Python · R · Bash · SQL

NGS AND AMPLICONS

QIIME 2 · DADA2 · cutadapt · VSEARCH · BLAST · decontam · MAFFT · FastTree

R

phyloseq · vegan · ANCOM-BC2 · glmmTMB · lme4 · dplyr · ggplot2 · R Markdown

PYTHON AND ML

pandas · NumPy · scikit-learn · matplotlib · seaborn · pytest · ensemble learning · feature selection

ENVIRONMENTS

Linux · HPC/SLURM · conda · Docker · Git · GitHub · GitLab

CURRENTLY LEARNING

Nextflow
Courses: Hello Nextflow and Nextflow Run (Seqera, 2026)

Profile

Bioinformatician with a PhD in Information Technology Research (USC, 2025), an MSc in Bioinformatics and Biostatistics and a degree in Computer Engineering. I work on NGS data analysis, mainly 16S rRNA amplicon sequencing and clinical metagenomics, building end-to-end analyses (quality control, preprocessing, statistical analysis and reporting) in Python, R and Bash on Linux and HPC environments with SLURM. I have developed open-source bioinformatics tools and applied machine learning to microbiome data. Six scientific publications, three as first author in Q1 journals.

Experience

Collaborating Researcher — Sep 2026 – Present
Universidade de Santiago de Compostela — Santiago de Compostela

- Processing of 16S rRNA amplicon sequencing data (V3-V4, Illumina paired-end) from clinical oral microbiome studies, combining several sequencing runs: quality control, primer trimming (cutadapt), per-run DADA2 denoising, taxonomic assignment (VSEARCH), contaminant removal with negative controls (decontam) and phylogeny (MAFFT, FastTree) in QIIME 2, with version-controlled scripts.
- Statistical analysis in R (phyloseq, vegan, ANCOM-BC2, glmmTMB): alpha and beta diversity, PERMANOVA, differential abundance and mixed models for paired designs.
- Methods and results reports (R Markdown and LaTeX) and documented deliverables for clinical research teams.

AI Research Technician — Jun 2025 – Jun 2026
CESGA, Galicia Supercomputing Center — Santiago de Compostela

- Developed AIRA16S during a research stay at CiTIUS (USC): a Python pipeline for the automated inventory of human 16S rRNA sequencing projects in public databases (NCBI SRA, BioProject, PubMed, Europe PMC), with LLM-based metadata extraction run on an HPC cluster, semantic validation and an interactive results viewer.
- OneHealth DataSpace project (NextGenerationEU), following Scrum: first version of the project website (Next.js, TypeScript, Git) and support for the deployment of the data space platform.

Predoctoral Researcher — Oct 2020 – May 2025
USC, CiTIUS · Xunta de Galicia grant ED481A-2021 — Santiago de Compostela

- PhD thesis (BIOMETA) on bioinformatics procedures for microbiome analysis through 16S rRNA sequencing, applied to the oral microbiome, with model training and execution on an HPC cluster via SLURM.
- Developed three open-source tools, version-controlled with Git and published on GitLab, each with a first-author Q1 publication: PrimerEvalPy, 16SGOSeq and EPheClass.
- International research stay at HEIG-VD, Switzerland (2022), where I developed the first version of EPheClass.

Software Developer — Mar 2019 – Sep 2020
Altia — Santiago de Compostela
Software development in an industry environment, first as an intern.

Software

- **PrimerEvalPy**: *in silico* evaluation of primers and primer pairs against sequence databases, with coverage metrics at each taxonomic level. [GitLab]
- **16SGOSeq**: curated database of 16S rRNA genes and their variants in complete genomes of oral bacteria and archaea, with taxonomy from domain to strain. [GitLab]

PROFILES

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Google Scholar

- **EPheClass**: phenotype classification from 16S rRNA count tables with a dynamic ensemble of machine learning models (kNN, Random Forest, SVM, XGBoost, MLP); ROC AUC 0.974. [GitLab]
- **Oral16SAmpliconSimilarity**: *in silico* analysis of 16S rRNA amplicon similarity across oral prokaryotic genomes, to identify which species become indistinguishable depending on the primer pair used. [GitLab]
- **AIRA16S**: automated inventory of human 16S rRNA sequencing projects in public databases, with LLM-based metadata extraction and semantic validation, integrated into a health data space.

Publications

- Vázquez-González, L. et al. (2024). **PrimerEvalPy: a tool for in-silico evaluation of primers for targeting the microbiome.** *BMC Bioinformatics (Q1)*. doi:10.1186/s12859-024-05805-7
- Vázquez-González, L. et al. (2025). **A curated bacterial and archaeal 16S rRNA Gene Oral Sequences dataset.** *Scientific Data (Q1)*. doi:10.1038/s41597-025-05050-4
- Vázquez-González, L. et al. (2025). **EPheClass: ensemble-based phenotype classifier from 16S rRNA gene sequences.** *Frontiers in Bioinformatics (Q1)*. doi:10.3389/fbinf.2025.1514880
- Regueira-Iglesias, A., Vázquez-González, L. et al. (2023). **In silico evaluation and selection of the best 16S rRNA gene primers for use in next-generation sequencing to detect oral bacteria and archaea.** *Microbiome (Q1)*. doi:10.1186/s40168-023-01481-6
- Regueira-Iglesias, A., Vázquez-González, L. et al. (2023). **Impact of 16S rRNA Gene Redundancy and Primer Pair Selection on the Quantification and Classification of Oral Microbiota in Next-Generation Sequencing.** *Microbiology Spectrum (Q2)*. doi:10.1128/spectrum.04398-22
- Regueira-Iglesias, A., Vázquez-González, L. et al. (2022). **In-Silico Detection of Oral Prokaryotic Species With Highly Similar 16S rRNA Sequence Segments Using Different Primer Pairs.** *Frontiers in Cellular and Infection Microbiology (Q1)*. doi:10.3389/fcimb.2021.770668

Conferences

- Vázquez-González, L. et al. (2023). **An Ensemble-Based Phenotype Classifier to Diagnose Crohn's Disease from 16S rRNA Gene Sequences.** *11th Iberian Conference on Pattern Recognition and Image Analysis (IbPRIA), Alicante. Poster.* doi:10.1007/978-3-031-36616-1_44
- Vázquez-González, L. et al. (2021). **PrimerEvalPy: a tool for in-silico evaluation of primers for targeting the microbiome.** *43rd Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC), online. One-page paper.*